

# One-Step Equations

AQA · KS3 Year 8

Name: \_\_\_\_\_ Class: \_\_\_\_\_ Date: \_\_\_\_\_

Solve a linear equation that needs a single step. Use the inverse operation to undo what has been done to the unknown, doing the same to both sides to keep the equation balanced.

## VISUAL EXAMPLE

$$2x + 3 = 11$$

$$-3 \text{ both sides} \quad 2x = 8$$

$$\div 2 \text{ both sides} \quad x = 4$$



*An equation balances: do the same to both sides to keep it level.*

## COMMON MISTAKES

Only changing one side — whatever you do to one side you must do to the other.

Using the wrong inverse: the inverse of  $\times 3$  is  $\div 3$ , not  $- 3$ .

For  $x \div 4 = 3$  you multiply both sides by 4; do not divide.

Sign slips with negatives — for example,  $x + 7 = 2$  gives  $x = -5$ , not 5.

## METHOD

- 1 Look at what has been done to the unknown letter.
- 2 Choose the inverse operation that undoes it.
- 3 Apply that inverse to BOTH sides of the equation.
- 4 Simplify each side to get the letter on its own.
- 5 Check by substituting your answer back into the original equation.

## WORKED EXAMPLE

Solve:  $x + 7 = 12$

Step 1 — 7 has been added to  $x$ .

Step 2 — The inverse of  $+ 7$  is  $- 7$ .

Step 3 —  $x + 7 - 7 = 12 - 7$

Step 4 —  $x = 5$

Step 5 — Check:  $5 + 7 = 12$

1 Solve.

(2 marks)

a)  $x + 5 = 12$

b)  $y - 3 = 10$

c)  $m + 8 = 8$

#### HOW TO START

A number has been added to the letter — undo it by subtracting the same number from both sides.

For  $x + 5 = 12$ , subtract 5 from each side:  $x = 12 - 5$ .

#### ANSWERS

(a) = \_\_\_\_\_

(b) = \_\_\_\_\_

(c) = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

2 Solve.

(2 marks)

a)  $4x = 20$

b)  $x \div 3 = 6$

c)  $7x = 56$

#### HOW TO START

Here the letter has been multiplied or divided.

For  $4x = 20$ , the inverse of  $\times 4$  is  $\div 4$ :  $x = 20 \div 4$ .

For  $x \div 3 = 6$ , the inverse of  $\div 3$  is  $\times 3$ :  $x = 6 \times 3$ .

#### ANSWERS

(a) = \_\_\_\_\_

(b) = \_\_\_\_\_

(c) = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

**3** Solve.

(2 marks)

a)  $x - 9 = -2$

b)  $x + 11 = 4$

c)  $6x = -30$

**ANSWERS**

**(a)** = \_\_\_\_\_

**(b)** = \_\_\_\_\_

**(c)** = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

**4** Solve.

(3 marks)

a)  $x \div 5 = -4$

b)  $-3x = 21$

c)  $x + 2.5 = 6$

**ANSWERS**

**(a)** = \_\_\_\_\_

**(b)** = \_\_\_\_\_

**(c)** = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

**5** Solve. The letter is on the right this time.

(3 marks)

a)  $9 = x + 4$

b)  $20 = 5x$

c)  $-8 = x - 3$

**ANSWERS**

**(a)**     = \_\_\_\_\_

**(b)**     = \_\_\_\_\_

**(c)**     = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

**6** Form an equation and solve it.

(3 marks)

a) A number  $n$  is multiplied by 4 to give 28. Find  $n$ .

b) When 5 is subtracted from a number  $p$ , the result is 12. Find  $p$ .

**ANSWERS**

**(a)** = \_\_\_\_\_

**(b)** = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

(3 marks)

- Write an equation in  $w$ . (1 mark)
- Solve it to find the width  $w$ . (1 mark)
- Work out the perimeter of the rectangle. (1 mark)

(a) = \_\_\_\_\_

(b) = \_\_\_\_\_

(c) = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting



- 8** A delivery company charges the same fixed price £x for each parcel.  
Four parcels cost £26 in total.

(5 marks)

- a) Write an equation for the total cost. (1 mark)  
b) Solve it to find the price of one parcel. (2 marks)  
c) A customer has a £40 budget. What is the greatest number of whole parcels they can send? Show your working. (2 marks)

**ANSWERS**

- (a)** = \_\_\_\_\_  
**(b)** = \_\_\_\_\_  
**(c)** = \_\_\_\_\_

**METHOD** 1 See what was done to the letter | 2 Choose the inverse operation | 3 Do it to BOTH sides | 4 Simplify to leave the letter alone | 5 Check by substituting

## Self-reflection

---

How confident do you feel with this topic now? (tick one)

Not yet

Getting there

Confident

Really confident

Which question did you find the hardest, and why?

---

---

How do I decide which inverse operation to use?

---

---

One thing I will practise before the next lesson:

---

---

## Teacher key

---

Q1: a)  $x = 7$  b)  $y = 13$  c)  $m = 0$

---

Q2: a)  $x = 5$  b)  $x = 18$  c)  $x = 8$

---

Q3: a)  $x = 7$  b)  $x = -7$  c)  $x = -5$

---

Q4: a)  $x = -20$  b)  $x = -7$  c)  $x = 3.5$

---

Q5: a)  $x = 5$  b)  $x = 4$  c)  $x = -5$

---

Q6: a)  $4n = 28$     $n = 7$  b)  $p - 5 = 12$     $p = 17$

---

Q7: a)  $5w = 35$  b)  $w = 7$  cm c) Perimeter =  $2(7 + 35) = 84$  cm

---

Q8: a)  $4x = 26$  b)  $x = £6.50$  c)  $40 \div 6.50 = 6.15...$    6 whole parcels ( $6 \times £6.50 = £39$    £40)

---